

Inside A Compás

A flamenco metronome that is a web app wearing a native shell. Thirty-four packages build it; roughly half of them reach the user. Here is what each one is for.

Quasar 2.28.0 **Vue** 3.5.42 **Capacitor** 7.6.8 **Tone.js** 15.1.22 **TypeScript** 5.9.3

Every pixel you see is a web page. It runs in a browser tab on `acompas.org`, inside a `WKWebView` on iOS, and inside Chromium on the desktop — the same code in all three. What changes is the container.

That is worth holding onto, because it explains the shape of the dependency list: one set of packages builds a web application, a second wraps it in something installable, and a third never ships at all.

What you see

The interface, its language, and its motion.

SHIPS TO USERS

vue 3.5.42

The framework everything else is built on. You describe what the screen should look like for a given state; Vue keeps the DOM matching it.

quasar 2.28.0

A component library and design system for Vue — buttons, dialogs, drawers, toggles, dark mode, the responsive grid. It is why the app looks consistent without hand-written CSS for every control.

IN THIS APP

Three of its plugins are switched on: `Notify` (the toast that tells you to select an instrument), `Loading`, and `Dialog`.

pug 3.0.4

An indentation-based template language used instead of HTML. Every `.vue` file here declares `<template lang="pug">`, which is why the markup has no closing tags.

IN THIS APP

Compiles to plain HTML at build time, so it costs the user nothing.

vue-router 5.3.0

Maps URLs to screens. Each page has its own address.

IN THIS APP

Uses real URLs on the web and hash URLs (`#/flamenco/solea`) in the native builds, because there is no server to ask for a path when the page is loaded from inside the app bundle.

vue-i18n 9.14.5

Translation. Text lives in per-language files rather than in components.

IN THIS APP

Ten languages, including Arabic and Persian, which read right to left. There is deliberately no i18n build plugin: the messages here are `.ts` modules, not the JSON/YAML resources such plugins precompile, and the current one defaults to a vue-i18n build with no message compiler — which silently emptied every translated control while the build still succeeded.

Translation. Text lives in per-language files rather than in components.

IN THIS APP

Ten languages, including Arabic and Persian, which read right to left.

`@intlify/vite-plugin-vue-i18n` compiles those files at build time so translation lookups are fast at runtime.

animejs 3.2.2

A small animation library. Given a target and a property, it interpolates over time.

IN THIS APP

The pulse on each dot as its beat lands — scale up, ease back down.

@quasar/extras 1.18.0

Icon sets and fonts packaged for Quasar. Only `mdi-v7` (Material Design Icons) is enabled; the other five are commented out so they are never downloaded.

vue-global-events

3.0.1

Lets a component listen for keyboard events on the whole window without manually adding and removing listeners. Powers the keyboard shortcuts.

What it remembers

Shared state, and the parts that survive a restart.

SHIPS TO USERS

pinia 2.3.1

The state store. Rather than passing data down through components, everything reads from a central place and updates flow outward.

IN THIS APP

Two stores. `patterns` holds the selected palo, tempo, instruments and beat events; `session` holds preferences like dark mode and the audio-visual offset.

@vueuse/core 10.11.1

A large collection of small Vue utilities.

IN THIS APP

Almost entirely one function: `useStorage`, which mirrors a value into `localStorage` automatically. It is why your tempo, language and chosen pattern are still there next time.

What makes the sound

The hard part, and the reason a metronome is not just a timer.

SHIPS TO USERS

tone 15.1.22

A framework over the browser's Web Audio API. Its value is scheduling: JavaScript timers drift audibly, so Tone schedules notes against the audio hardware clock instead, looking ahead and queueing events before they are due.

IN THIS APP

`Transport` is the clock that tempo and swing act on. `Sequence` plays one pattern array per instrument. `Player` holds each decoded sample, `Channel` mixes them, `Reverb` adds room, and `Draw` schedules the dot animations against the same clock so sight and sound stay together.

The written material

Help pages, palo descriptions, the changelog.

SHIPS TO USERS

marked 14.1.4

Turns Markdown into HTML. The help text and palo documentation are authored as Markdown inside the translation files.

dompurify 3.4.14

Strips anything dangerous out of HTML before it is inserted into the page. Standard practice whenever generated HTML is injected — it closes off cross-site scripting.

IN THIS APP

Paired with `marked` : parse, then sanitize, then render. Some of that text is fetched from Wikipedia, so it is not all locally authored.

The native shell

What turns a web page into something you install.

SHIPS TO USERS

@capacitor/core 7.6.8

Wraps the web app in a real native application: a Swift or Kotlin project containing a full-screen web view, plus a bridge that lets JavaScript call native code. The web assets are copied inside the app bundle and served through a custom URL scheme, so the app works with no network at all.

IN THIS APP

That scheme handler is also where the iOS audio bug lived — it answers requests for media files in a way that `fetch()` reads as a failure.

@capacitor/app · splash-screen · status-bar

7.1.2 / 7.0.5 / 7.0.6

Small official plugins exposing native behaviour to JavaScript: lifecycle events such as pause and resume, the launch splash, and the colour and style of the status bar.

@capacitor-community/keep-awake 7.1.0

Stops the screen sleeping. For a metronome propped on a music stand, this is not a nicety.

electron 31.7.7

The desktop equivalent: bundles Chromium and Node.js into a Mac or Linux application. Heavier than Capacitor — the browser engine ships with the app rather than being borrowed from the operating system.

IN THIS APP

`@electron/remote` lets the page reach a few main-process APIs;
`electron-builder` produces the installers.

What builds it

Runs on your machine. None of this reaches a user.

NEVER SHIPS

@quasar/app-vite 3.8.1

The build system and CLI. It owns `quasar dev` and `quasar build`, wires Vue, the router, the store and Quasar together, and knows how to target each platform — the reason `-m capacitor -T ios` means anything.

WORTH KNOWING

It carries its own Vite (8.2.2 here) rather than taking it as a dependency. The project moved v1 → v2 → v3 in one sitting; each hop broke the config differently, and both of the interesting failures compiled cleanly before breaking at runtime.

typescript 5.9.3

JavaScript with type annotations, checked at build time and then erased. The pattern and sound data are typed, which is what makes a malformed sequence a build error rather than silence at runtime.

eslint · prettier 8.57.1 / 3.9.6

ESLint catches suspect code; Prettier formats it. `eslint-config-prettier` switches off the ESLint rules that would argue with Prettier, and `eslint-plugin-vue` teaches ESLint about `.vue` files.

autoprefixer 10.5.4

Adds vendor prefixes to CSS for the browsers listed in the build target, so you write the standard property once.

electron-builder 24.13.3

Packages the Electron app into a `.dmg` or Linux package, handling signing and metadata. Selected in `quasar.config.js` via `bundler: 'builder'`.

@capacitor/cli 7.6.8

Copies built web assets into the native projects and keeps their plugin lists in sync. Quasar calls it for you during a Capacitor build.

Declared but not carrying weight

A grep for package names finds nothing for a third of the dependency list. Some of those were genuinely dead and have been removed; the rest are invoked as tooling or referenced by a name that is not the package path.

PACKAGE

WHY IT IS HERE

removed

Six packages were dropped on 2026-08-30: `pug-plain-loader` (a webpack loader in a Vite project), `electron-packager` (the config selects `builder`; deprecated upstream), `@types/dompurify` (DOMPurify 3 ships its own types), `@vue/devtools` (app-vite v3 uses `vite-plugin-vue-devtools`, installed on demand), `@capacitor/assets` (icons come from `icongenie.sh`) and `@vue/language-plugin-pug` (never wired into `vueCompilerOptions`).

@types/animejs

Types for anime.js, which ships none of its own — genuinely load-bearing despite nothing importing it by name. Kept.

PACKAGE

WHY IT IS HERE

@quasar/extras

Named in `quasar.config.js` as `'mdi-v7'` rather than by package path, so a search for the package name misses it.

eslint · typescript · autoprefixer · electron-builder · @capacitor/cli

Invoked as tooling by the build or by ESLint's own config; never imported from source.

Versions shown are what is installed in `node_modules`, which can sit ahead of the range declared in `package.json` — `dompurify` is declared as `^3.1.6` and installed at `3.2.6`.

The audio samples are not a package. They live in `public/audio` as 321 recordings in five formats, generated from WAV masters by `format_audio.py`, and account for roughly 99 MB of the iOS app.